

Reconstruction of long-lived particles with the CLUE algorithm in CMS

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Abstract

The CMS experiment at the LHC is upgrading its endcap calorimeters with the High-Granularity Calorimeter (HGCAL). Long-lived particles can decay far from the interaction point and produce photons whose trajectories do not point back to it. Reconstructing these photons is a challenge for CLUE3D, the clustering algorithm developed for HGCAL.

This project studies CLUE3D performance using simulated single- and two-photon events, with and without pileup. We improved the displaced-photon generator and added displacement observables to the standard validation tools. The results show that CLUE3D loses efficiency as displacement increases, with single showers fragmenting and nearby showers becoming mixed. Adding pileup leaves the signal-photon efficiency largely unchanged. Tests of the alternative CLUEstoring algorithm show that suitable configurations improve displaced-photon reconstruction while keeping nearby showers separate. Further tuning and tests with realistic long-lived-particle decays are needed.

1 Introduction

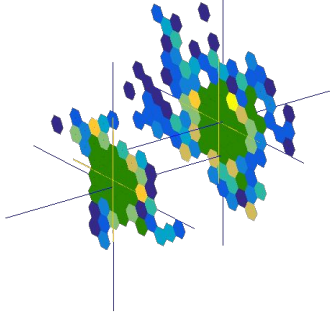
The year 2030 will mark the start of the High Luminosity phase of the Large Hadron Collider (LHC) [1], which will deliver proton–proton collisions at an unprecedented luminosity. The resulting data volume and the severe pileup—of order 200 simultaneous interactions per bunch crossing—demand a new generation of reconstruction algorithms. To cope with these conditions the CMS experiment will replace its endcap calorimeters with the High-Granularity Calorimeter (HGCAL) [2], a sampling calorimeter with roughly six million channels arranged in 47 longitudinal layers. Its fine transverse and longitudinal segmentation capture detailed three-dimensional structure of electromagnetic and hadronic showers, while also providing energy capture timing information.

The Clustering of Energy (CLUE) algorithm [4] is a fast, parallel, energy density-based clustering algorithm that groups HGCAL energy deposits within each layer. In the CMS reconstruction, the Iterative Clustering (TICL) framework [5] builds objects in stages (Fig. 1): individual calorimeter hits (*RecHits*) are first grouped per layer by CLUE into two-dimensional *LayerClusters* (LCs). A three-dimensional extension of CLUE, CLUE3D, then links *LayerClusters* across multiple layers into *Tracksters*; a subsequent linking step associates *Tracksters* into the final TICL candidates. The Monte Carlo truth is represented by *SimTracksters*, which collect the *LayerClusters* associated with a simulated shower. Reconstruction performance is measured by comparing reconstructed *Tracksters* to *SimTracksters* through their shared energy.

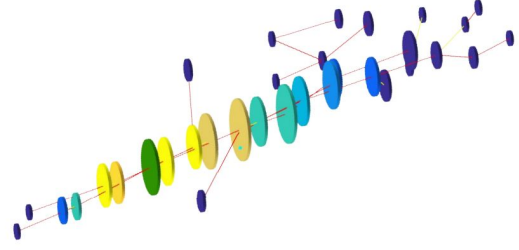
An important CLUE3D assumption: the neighbour search that links layer clusters across layers is performed in a coordinate system that projects layers towards the interaction point (IP). Consequently, a shower travelling outward from the CMS centre has a fixed projected position from layer to layer. This holds for *prompt* particles produced at the IP, but it doesn't need to for the class of *long-lived particles* (LLPs) predicted by many extensions of the Standard Model [6]. Rather than decaying within a few millimetres of IP, an LLP can travel longer distance in the detector before decaying, producing *displaced*, non-pointing particles. A representative signature is a dark shower in which long-lived dark hadrons decay to Standard Model photons far from the beamline. Because these showers do not point back to the origin, they may violate exactly the assumption on which CLUE3D relies.

The goal of this project is to determine if CLUE3D is capable of clustering the energy deposits from such novel topologies, to measure the expected performance degradation as a function of the distance from IP to where LLP decays, and to explore an alternative algorithm. We restrict the study to *photons*: neutral particles whose trajectory is not curved by the CMS magnetic field, so they go in straight lines and their displacement is unambiguously defined. We also produce them immediately in front of HGCAL so they also do not interact in the tracker. This isolates the calorimeter clustering behaviour from tracking and magnetic-field effects and provides the cleanest possible test of the origin assumption.

The remainder of this report is organized as follows. Section 2 defines the displacement observables for re-



(a) RecHits and LayerClusters



(b) LayerClusters linked into a Trackster

Figure 1: The HGCAL reconstruction chain. (a) Calibrated energy deposits (RecHits) of two photon showers. CLUE groups them by layer into two-dimensional LayerClusters. (b) CLUE3D links LayerClusters across layers into a three-dimensional Trackster; the red lines are the links that build the shower. Images from Ref. [3].

constructed objects, while Sections 3 and 4 describe the displaced-photon Monte Carlo production and performance metrics. Section 5 reviews how CLUE3D works, with emphasis on the origin assumption. We then present the single-photon efficiency (Section 6), two-shower merging behaviour (Section 7), and performance under pileup (Section 8), before comparing CLUE3D with the alternative algorithm CLUEsterng (Section 9) and concluding in Section 10.

2 Displacement observables

Every result in this report is expressed as a function of how *displaced* a shower is. We need to define displacement observables and explain how they are calculated for reconstructed objects that carry no truth information.

2.1 Definitions

Consider a photon that is created at $\mathbf{x}_v = (x_v, y_v, z_v)$ and travels in a straight line to the HGCAL front face, located at $z \simeq 321$ cm from the interaction point (Fig. 2). HGCAL has two endcaps; in this report, we consider the positive- z endcap. Its layout is illustrated in Fig. 3. We measure the photon's displacement by two quantities:

- the **displacement radius** R , defined as the transverse distance from beamline at which the photon's extrapolated trajectory crosses $z = 0$ plane,

$$R = \sqrt{x_0^2 + y_0^2}, \quad (1)$$

where (x_0, y_0) is extrapolated trajectory to $z = 0$;

- the **displacement angle** α , the angle between the photon's momentum direction $\hat{\mathbf{p}}$ and the unit vector $\hat{\mathbf{r}}_{\text{face}}$ from the IP to its entry point on the HGCAL front face,

$$\alpha = \arccos(\hat{\mathbf{r}}_{\text{face}} \cdot \hat{\mathbf{p}}). \quad (2)$$

A prompt photon has $\hat{\mathbf{r}}_{\text{face}} \parallel \hat{\mathbf{p}}$ and $R = \alpha = 0$; the non-pointing photons studied here have $R > 0$ and $\alpha > 0$.

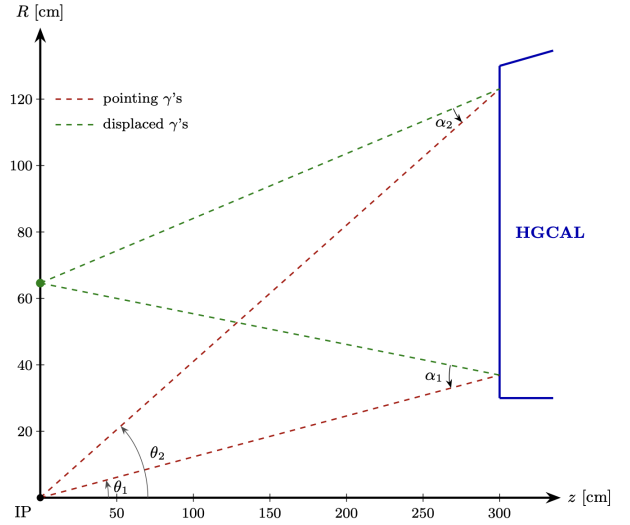


Figure 2: Displacement geometry in the (z, R) plane. Pointing photons (red) travel radially from the interaction point with $\alpha = 0$; displaced photons (green) originate at a transverse distance R from the beamline at $z = 0$ and hit the calorimeter with $\alpha > 0$. The two green trajectories show that a single R can give different angles α . Figure from Ref. [7].

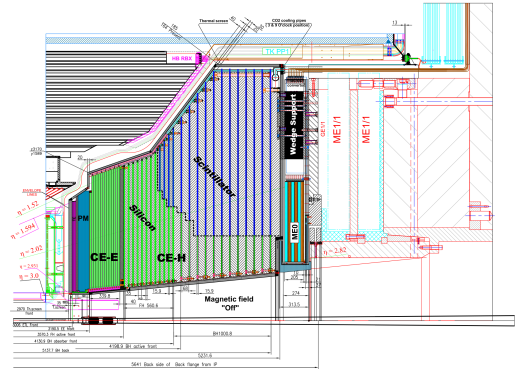


Figure 3: Longitudinal cross section of the upper half of one HGCAL endcap, showing the electromagnetic (CE-E) and hadronic (CE-H) compartments. Figure from Ref. [2], Fig. 1.6.

For a truth photon these quantities are trivial, since the production vertex and momentum are known directly from the generator truth.

2.2 Reconstructing the displacement

The difficulty is on the reconstruction side. A reconstructed Trackster has no persisted vertex where it was produced, and it does not need to correspond one-to-one to a SimTrackster: it may contain only a fragment of a true shower, or it may merge multiple showers. The merge-rate metrics defined below require a displacement value. We therefore reconstruct R directly from the reconstructed Trackster geometry.

Each Trackster provides an energy-weighted barycentre $\mathbf{x}_c = (x_c, y_c, z_c)$ and a principal-axis direction $\mathbf{u} = (u_x, u_y, u_z)$ calculated by principal-component analysis (PCA) of its LayerClusters. Treating the shower as a straight line through $\mathbf{x}_c$ with direction $\mathbf{u}$ and extrapolating to $z = 0$ gives

$$x_0 = x_c - z_c \frac{u_x}{u_z}, \quad y_0 = y_c - z_c \frac{u_y}{u_z}, \quad (3)$$

and $R_{\text{reco}} = \sqrt{x_0^2 + y_0^2}$ from Eq. (1). This straight-line treatment is correct for photons, which makes them the ideal probe for validating performance.

2.3 Estimate error and the choice of direction estimator

The accuracy of R_{reco} sets resolution for reco-based metrics, so we measured its error,

$$\Delta R \equiv R_{\text{reco}} - R_{\text{truth}}, \quad (4)$$

summarized by its RMS. Starting from the ordinary PCA that CMSSW already exposes, the error is broad, $\text{RMS}(\Delta R) \approx 20$ cm. This error has only two possible sources, the barycentre and the direction, and separating them shows that it comes almost entirely from the *direction*: reconstructing only barycentre and using the true direction gives 1.1 cm, and reconstructing only direction and using the true barycentre gives ~ 20 cm. It is because the barycentre sits inside the shower and is well measured, but the small angular uncertainty of the direction is amplified over $z_c \sim 320$ cm extrapolation back to $z = 0$.

Since the direction estimate dominates the error, we tested several direction estimators:

- **Ordinary PCA:** the CMSSW default, using all LayerClusters.
- **Clean PCA:** PCA using only the leading LayerCluster per layer in a window 10 layers before and around the shower maximum.
- **Endpoint layers:** the direction between the first and last populated layers in the window.
- **Window barycentres:** an axis constructed from per-layer energy-weighted barycentres.

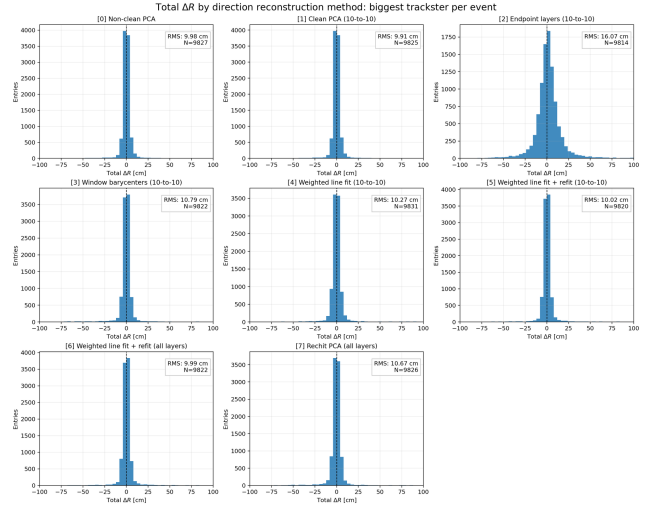


Figure 4: Error distribution ΔR for each direction estimator, restricted to the leading Trackster per event. The PCA variants are the narrowest (clean PCA 9.91 cm, ordinary 9.98 cm).

- **Weighted straight-line fit:** an energy-weighted fit to the LayerClusters.
- **Weighted line fit with outlier refit:** the same fit after removing the LayerClusters with the largest residuals.
- **RecHit PCA:** PCA performed on raw RecHits rather than LayerClusters.

Figure 4 compares these methods for the leading Trackster per event. The PCA variants are already the narrowest; because clean PCA is robust and implemented in CMSSW, it was adopted. Window barycentres may be faster to compute than PCA, but runtime benchmarks would be needed to compare the methods at similar spatial resolution.

The resolution improves with Trackster energy, as a more energetic shower gives the PCA more LayerClusters to fit its axis; ~ 10 cm is leading-Trackster resolution.

The displacement observables R and α , together with a Trackster-timing observable, were added to the standard HGCAL validation as new per-object quantities so that any efficiency, purity, duplicate, fake or merge plot can be produced as a function of displacement (CMSSW pull request #51510 [8]).

3 Displaced photon gun

As mentioned before, we chose to study displaced photons due to their shower simplicity and straight trajectory. The original displaced-particle gun was already developed for the project (CMSSW pull request #51006); my contribution was to test and improve it—modifying its sampling and configuration, correcting the production timing, and updating the beamspot choice—done in pull request #51489 [9].

Photons are produced immediately in front of HGCAL, so they do not convert or shower in the tracker.

This isolates energy response to HGCAL. The gun is a controlled test of the reconstruction geometry, not a simulation of a full LLP production and decay.

3.1 Trajectory geometry

The gun samples the origin in the plane $z = 0$,

$$\mathbf{x}_0 = (R \cos \phi_0, R \sin \phi_0, 0), \quad (5)$$

so its radius is exactly the displacement observable of Section 2. A straight trajectory of polar angle θ and azimuth ϕ_p starts from this origin, and the photon is created where the line crosses a *production plane* near the HGCAL front face ($z \simeq 319$ cm); an optional *target plane* near the back of the electromagnetic calorimeter can constrain a second annular acceptance so that the trajectory remains within the calorimeter volume. This helps ensure transverse shower containment, so measurements of reconstruction performance are not affected by energy leakage. Longitudinal containment is ensured by the calorimeter depth in radiation lengths.

To keep the requested displacement distribution unbiased, the gun holds the sampled origin fixed and after choosing R it constructs the allowed polar-angle intervals analytically before sampling θ . It samples R uniformly, giving equal populations per displacement bin, and offers either a uniform total-energy or a uniform transverse-momentum distribution. Sampling E directly keeps the energy distribution independent of the accepted direction, which is useful when isolating displacement effects.

3.2 Timing, energy and beamspot corrections

Validating the pointing limit $R = 0$ against the existing CloseBy gun exposed three issues, all now fixed. First, the original gun assigned $t = 0$ to a photon created in front of HGCAL, ignoring the flight time required to reach that point. Its signals therefore arrived about 11 ns earlier than a particle travelling at the speed of light from the IP could reach the calorimeter. These arrival times are incompatible with the timing assumptions of hit reconstruction, so such signals could be rejected. The resulting reconstructed energy was reduced and fragmented across the shower, producing several Tracksters even at $R = 0$. This was therefore a consequence of the incorrect generation time, rather than a displacement-caused clustering failure. The corrected version assigns the physical flight time,

$$ct_{\text{prod}} = |\mathbf{x}_0| + \frac{|\mathbf{x}_{\text{prod}} - \mathbf{x}_0|}{\beta}, \quad \beta = \frac{|\mathbf{p}|}{E}. \quad (6)$$

For photons, $\beta = 1$, so at $R = 0$ this reduces to the ordinary ~ 11 ns flight time to the production plane. Second, the gun sampled p_T rather than energy, which produced much higher energies at small momentum θ ($E \propto p_T / \sin \theta$); direct energy sampling was added. Third, the workflow used the DBRealisticHLLHC beamspot

smearing and was switched to the fixed CloseBy production. Together these corrections reproduce the CloseBy response at $R = 0$.

4 Reconstruction performance metrics

We will study how shower energy is distributed among reconstructed objects and contamination when several showers overlap. We use the HGCAL validation framework, which associates reconstructed Tracksters to SimTracksters by shared LayerCluster energy and from that defines efficiency, duplicate and merge rates.

The association compares the per-LayerCluster energies of a SimTrackster S and a reconstructed Trackster T . The shared energy $E_{\text{shared}}(S, T)$ is the energy common to both, and the recovered fraction $f_{ST} = E_{\text{shared}}(S, T)/E_S$ is the fraction of the true shower energy E_S in a single reconstructed object. Two directional scores, $s_{S \rightarrow T}$ and $s_{T \rightarrow S}$ (weighting squared energies, lower being better), separately penalize SimTrackster energy missing from T and reconstructed energy in T missing in S .

4.1 Efficiency, duplicate rate and merge rate

For efficiency, a reconstructed Trackster T is matched to a SimTrackster S if T holds more than half of S 's energy, i.e. if $f_{ST} > 0.5$. A SimTrackster is counted as efficient if at least one reconstructed Trackster is matched to it. This criterion does not check whether that T contains energy from several SimTracksters. Binned in a truth observable B (e.g. displacement R), we define efficiency as

$$\epsilon(B) = \frac{N_S(B; \exists T : f_{ST} > 0.5)}{N_S(B)}. \quad (7)$$

For the duplicate rate, T is matched to S if $s_{S \rightarrow T} < 0.6$; the rate counts SimTracksters that match more than one reconstructed Trackster. For the merge rate, S is matched to T if $s_{T \rightarrow S} < 0.6$; the rate is defined on reconstructed Tracksters, binned by reconstructed R , and counts T that have more than one S matching. The formal definitions are

$$D(B) = \frac{N_S(B; \#\{T : s_{S \rightarrow T} < 0.6\} \geq 2)}{N_S(B)}, \quad (8)$$

$$M(B) = \frac{N_T(B; \#\{S : s_{T \rightarrow S} < 0.6\} \geq 2)}{N_T(B)}. \quad (9)$$

Small fragments can fail this cut, so a low duplicate or merge rate does not by itself exclude fragmentation; it is a caveat revisited in the two-shower study.

4.2 Validation changes

Making the merge, duplicate and other score thresholds configurable, displacement and timing observables, and

Table 1: Default HLT CLUE3D parameters.

Parameter	Value
criticalDensity	0.6 GeV
densitySiblingLayers	3
densityXYDistanceSqr	3.24 cm ² ($d_c = 1.8$ cm)
kernelDensityFactor	0.2
criticalXYDistance	1.8 cm (d_m)
outlierMultiplier	2

a later refactoring that added two-dimensional metrics in R and a second variable, went into CMSSW pull requests #51510 [8] and #51565 [10].

5 How CLUE3D works

CLUE3D is the existing CMS algorithm that groups LayerClusters across calorimeter layers into Tracksters using the density-based idea of CLUE [4]. Its CMSSW implementation is available in `PatternRecognitionbyCLUE3D.cc` [11]. The studies use the default HLT CLUE3D configuration without tuning for displaced showers, except where a parameter variation is explicitly stated. Table 1 lists the parameters referenced below; CLUE3D has additional configuration parameters that are not discussed here. Values set by the HLT configuration override the algorithm defaults.

5.1 Neighbourhoods projected from the interaction point

Let LayerCluster i have position (x_i, y_i, z_i) , transverse radius $r_i = \sqrt{x_i^2 + y_i^2}$, azimuth ϕ_i , layer index ℓ_i , and energy E_i . To compare a candidate j with i , it projects j onto the plane of i along the ray from the origin through j . The projected radius is

$$r_{j \rightarrow i} = z_i \frac{r_j}{z_j}. \quad (10)$$

With the azimuthal difference $\Delta\phi_{ij}$ wrapped into $[-\pi, \pi]$, the radial and azimuthal components of the projected separation are

$$d_{r,ij} = r_i - r_{j \rightarrow i}, \quad d_{\phi,ij} = r_{j \rightarrow i} \Delta\phi_{ij}. \quad (11)$$

The squared projected distance in the transverse (x, y) plane, denoted d_{XY}^2 , is

$$\begin{aligned} d_{XY}^2(i, j) &= d_{r,ij}^2 + d_{\phi,ij}^2 \\ &= (r_i - r_{j \rightarrow i})^2 + r_{j \rightarrow i}^2 (\Delta\phi_{ij})^2, \end{aligned} \quad (12)$$

This expression uses the small-angle approximation to represent the azimuthal separation as an arc length. This is a projected distance, not Euclidean distance of the two LayerClusters. Exchanging i and j changes the reference plane and can change its value, so $d_{XY}(i, j) \neq d_{XY}(j, i)$.

Candidate searches are restricted to three layers before and after the reference layer. They also use position-based (η, ϕ) buckets called tiles. On each searched layer, LC candidates for density are taken from tiles within two bins in both η and ϕ of the reference LayerCluster's tile. The nearest-higher search uses one bin in each direction.

5.2 Density and Trackster formation

The local density of LayerCluster i is

$$\rho_i = E_i + k_\rho \sum_{j \in \mathcal{N}_i} E_j, \quad (13)$$

where $k_\rho = 0.2$ is the `kernelDensityFactor`, and $\mathcal{N}_i$ contains candidate LayerClusters on other layers with $d_{XY}(i, j) < d_c$, and $d_c = 1.8$ cm. The algorithm then finds the nearest higher-density LayerCluster $h(i)$ from a layer $\ell_{h(i)} \in [\ell_i - 3, \ell_i + 3]$, with $\ell_{h(i)} \neq \ell_i$, minimizing the projected transverse distance, and records $\delta_i = d_{XY}(i, h(i))$. This search can find neighbours beyond d_c ; if none is found, the distance is treated as infinite.

LayerClusters are classified as follows:

- **Seeds** start new Tracksters. They require $\rho_i \geq \rho_c = 0.6$ GeV and separation from nearest higher: $\delta_i > d_m = 1.8$ cm.
- **Outliers** have low density, $\rho_i < \rho_c$, and a distant nearest-higher neighbour, $\delta_i > 2d_m = 3.6$ cm.
- **Followers** are the remaining LayerClusters and inherit the Trackster assignment of their nearest-higher neighbour, if one exists.

Starting from each seed, its Trackster index is propagated through the follower links (Fig. 1b). Each assigned LayerCluster belongs fully to one Trackster. Outlier status propagates the same way: a follower whose nearest-higher chain terminates in an outlier instead of seed is itself treated as an outlier.

6 CLUE3D performance for displaced photons

The central question of this project is whether CLUE3D can reconstruct a photon shower whose trajectory does not point back to the interaction point. The displaced-photon study shows both a performance loss and an unexpected structure: the loss develops in two stages, separated by a plateau. The study shows that fragmentation induced by the origin-based projection is the mechanism behind the observed loss.

6.1 The performance loss is not a smooth decline

We used single displaced photons without pileup, with Run4D121¹ geometry and the default HLT CLUE3D configuration. The wide-range production samples R uniformly from 0 cm to 350 cm and photon energy from 10 GeV to 250 GeV, subject to the front and back HGCAL-EE surface constraints described in Section 3. We used 50 000 events, each containing one displaced photon.

¹The CMSSW identifier for the specific Phase-2 detector geometry used in this study.

In addition to the efficiency of Section 4, I recorded the largest fraction of each SimTrackster's energy captured by any one reconstructed Trackster,

$$f_{\max}(S) = \max_T \frac{E_{\text{shared}}(S, T)}{E_S}. \quad (14)$$

Efficiency is the fraction of SimTracksters with $f_{\max} > 0.5$. Mean $\langle f_{\max} \rangle$ measures energy recovery continuously, without reducing each shower to a pass or fail, and does not have a threshold.

Figure 5 shows that the loss is not a smooth decline. The efficiency is high at small displacement and drops towards the end, but in between there is a plateau, seen even more sharply in $\langle f_{\max} \rangle$.

This plateau is unexpected: increasing displacement should progressively violate the pointing assumption, yet reconstruction temporarily stops deteriorating. The feature persists whatever shared-energy requirement is used, so it is not a consequence of the 0.5 efficiency threshold. We will investigate real cause to efficiency loss and plateau.

6.2 Testing geometric explanations

Some explanations can be ruled out quickly. The first is the limited (η, ϕ) tile search: a displaced shower does not keep a constant (η, ϕ) from layer to layer, so its LayerClusters could drift out of the search window. But it was not the case, almost every LayerCluster falls in the same tile as the shower core, at most one tile away, well within the search reach, so no same-shower neighbour is lost to tiling. The second explanation is a hidden dependence on trajectory orientation which causes plateau. However, separating photons by $\Delta\phi(\phi_{\text{origin}}, \phi_{\text{surface}})$ left the plateau intact, so the mixing of orientations under a scalar R does not create the feature either.

6.3 The geometric origin of the growing distance

The projected distance of Section 5 suggests a mechanism. For a straight trajectory, transverse position is

$$\mathbf{r}(z) = \mathbf{b} + z\mathbf{a}, \quad R = |\mathbf{b}|, \quad (15)$$

where $\mathbf{a}$ is the transverse slope and $\mathbf{b}$ the position at $z = 0$. For a pointing trajectory $\mathbf{b} = 0$: both r and ϕ are constant along the ideal shower axis. For a displaced trajectory, $\mathbf{r}(z)/z = \mathbf{a} + \mathbf{b}/z$ changes with depth, as Fig. 6 demonstrates for a real shower.

Projecting a point on layer j to layer i along a ray from the interaction point gives

$$\mathbf{r}_{j \rightarrow i} = \frac{z_i}{z_j} \mathbf{r}(z_j). \quad (16)$$

Subtracting this projected position of j from the position of i gives:

$$\Delta \mathbf{r}_{ij} = \mathbf{r}(z_i) - \mathbf{r}_{j \rightarrow i} \quad (17)$$

$$= \mathbf{b} \left(1 - \frac{z_i}{z_j} \right) = \mathbf{b} \frac{z_j - z_i}{z_j}. \quad (18)$$

Thus the exact offset between ideal axis points is $|\Delta \mathbf{r}_{ij}| = R|z_j - z_i|/z_j$. For neighbouring layers, and using the small-angle transverse metric implemented in CLUE3D,

$$d_{XY}(i, j) \simeq R \frac{|\Delta z|}{z}. \quad (19)$$

Equation (19) explains that distance grows with both displacement and layer separation even for a perfectly straight photon. It also gives the approximate scale at which a link exceeds the critical distance,

$$R_{\text{crit}} \simeq \frac{d_m z}{|\Delta z|}. \quad (20)$$

This supplies a prediction to test against the observed regions.

A first attempt looked promising but proved incomplete. Comparing, for the most energetic LayerCluster on each layer, the offsets of pairs separated by $d = 1, 2, 3$ layers, the mean $d = 3$ radial ($r_i - r_{j \rightarrow i}$) offset reaches 1.8 cm near the first decline and the $d = 2$ offset near the second, suggesting that links are lost as displacement grows. This radial-only picture does not survive once the full projected distance is used: the azimuthal term of d_{XY} is not negligible, so the crossings shift.

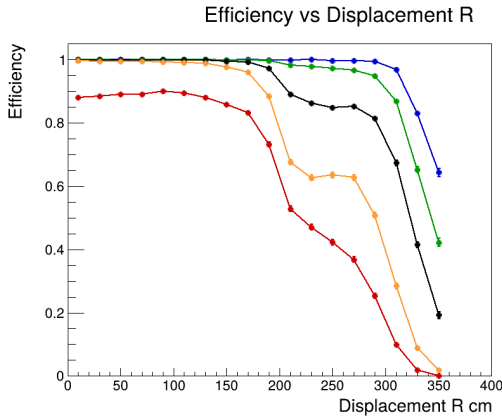
The local display in Fig. 7 shows why the full distance matters: the same-shower LC from another layer lies 1.82 cm from another seed in projected coordinates, just outside the critical circle, so two pieces of one continuous shower fall are not satisfying distance criterion.

6.4 The two-layer transition explains the first loss

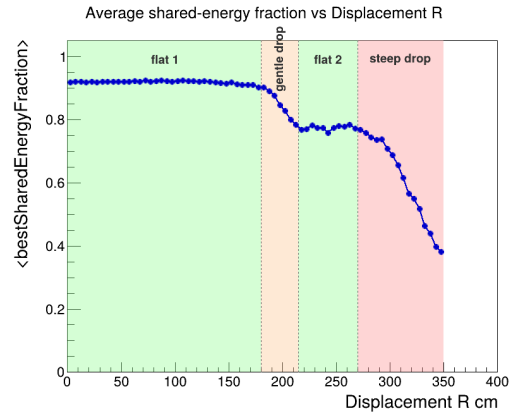
Important insight is shown in Fig. 8a. Near $R \simeq 200$ cm, the mean full projected distance for $d = 2$ pairs crosses 1.8 cm. At the same time, as shown in Fig. 8b, the energy carried by reco Tracksters which failed to follow nearest higher $d = 2$ layers away rises sharply. This connects the threshold to the objects that actually remove energy from the best reconstructed match, which reduces efficiency.

The $d = 2$ contribution rises from almost zero to roughly 15% of the shower energy through the first decline. It then has plateau over approximately the same displacement range as the plateau in $\langle f_{\max} \rangle$. The plateau therefore corresponds to a region where this newly important source of fragmentation has largely stabilized, before the next one becomes dominant. At larger displacement, the green $d = 1$ contribution rises rapidly and drives the second, stronger loss of captured energy.

Three-layer pairs cross 1.8 cm earliest, but remove little energy, because $d = 1, 2$ higher-density neighbour can still catch the LayerCluster and prevent a split. And although the mean one-layer distance stays below 1.8 cm for every R , individual one-layer links still increasingly exceed it and produce the observed contribution. This explains the first decline, the plateau and the second decline.



(a) Efficiency for several shared-energy requirements



(b) Mean largest shared-energy fraction

Figure 5: Single-photon reconstruction without pileup. (a) Efficiency for several shared-energy requirements; the black curve is the standard $f_{\max} > 0.5$. (b) Mean $f_{\max}$, showing the two-stage loss and the intermediate plateau. Shaded regions are approximate slope regions.

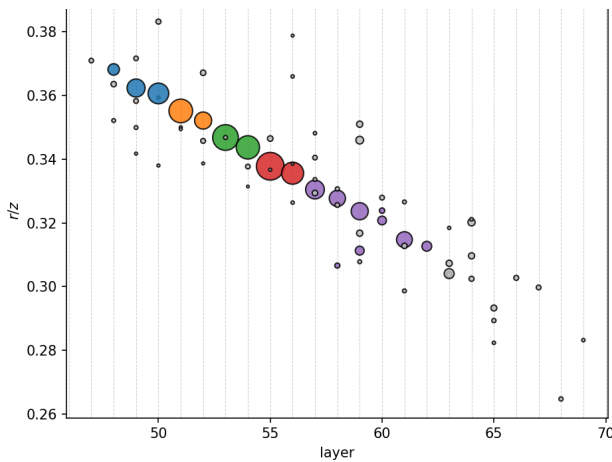


Figure 6: A single photon at $R \simeq 297$ cm split into five reconstructed Tracksters, shown as r/z versus layer index. Colour identifies the reconstructed Trackster and marker size the energy; grey markers are outliers. The layer index follows the TICL convention, in which positive-endcap layers are offset to accommodate the negative endcap. r/z is not constant.

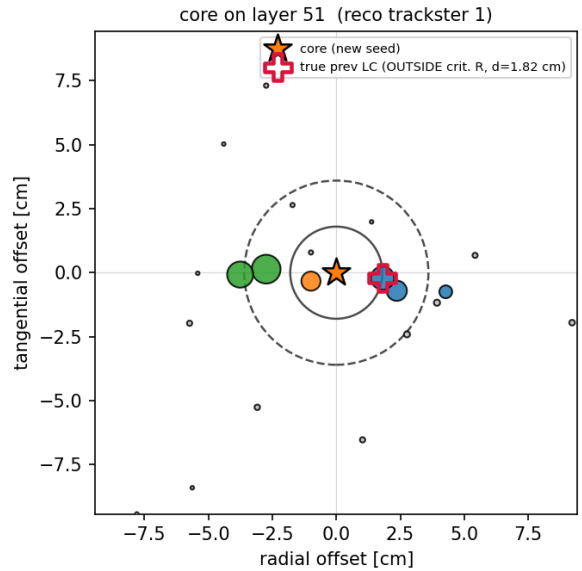


Figure 7: A neighbourhood in the full projected coordinates for the $R \simeq 297$ cm event previously shown in Fig. 6. The star marks the layer-51 core of a new fragment, and circles the 1.8 cm and 3.6 cm distances.

6.5 A controlled change of the critical distance

As a direct test, I reran the HLT reconstruction on the same sample with the seed distance raised from $d_m = 1.8$ cm to 2.1 cm, with the density radius fixed. By Eq. (20) this should push each fragmentation stage to larger R , and it does: the $d = 2$ seeded contribution turns on later, as shown in Fig. 9. The degradation moves where the geometry predicts, suggesting that it is caused by the projected-distance threshold and not just correlated with displacement.

7 Merging in two-photon events

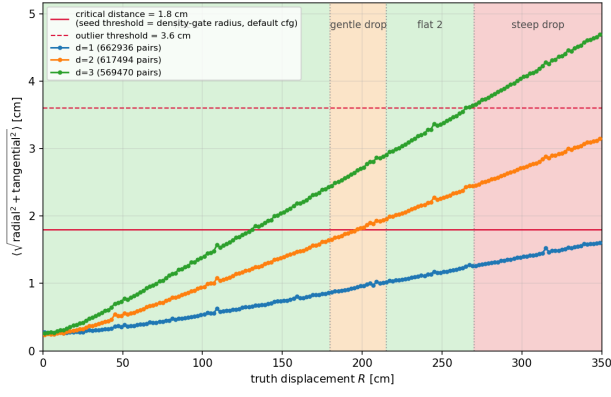
The single-photon study looked at fragmentation, but a useful reconstruction must also keep neighbouring showers separate. Increasing the nearestHigher distance can make displaced shower efficient, while

merging multiple of them. I therefore extended the displaced-gun studies to two photons. All results in this section use the default CLUE3D configuration.

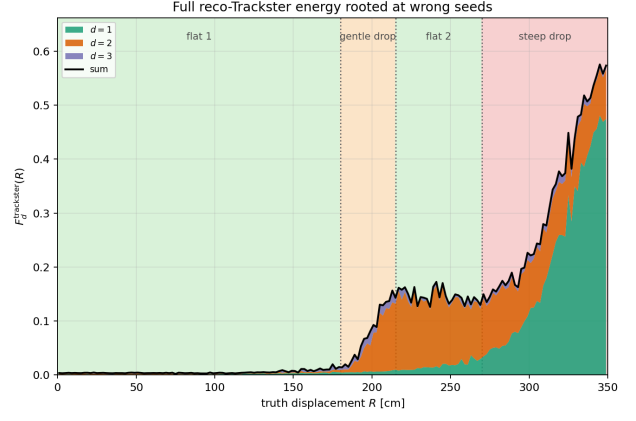
7.1 Controlled shower separation

The two photons are separated with a distance-scan generator: they have equal energies and parallel momenta, with production points separated by a distance $s = 4$ cm. This separation was chosen because the merging behaviour is most clearly visible there. Other values of s were also studied, as shown in Fig. 11, but are not discussed further. The study uses 50 000 generated events.

Merge rate initially appears encouraging at large R : after rising at intermediate displacement, it falls again (Fig. 10a). However, this is the same region in which single showers fragment strongly. Fragmentation can

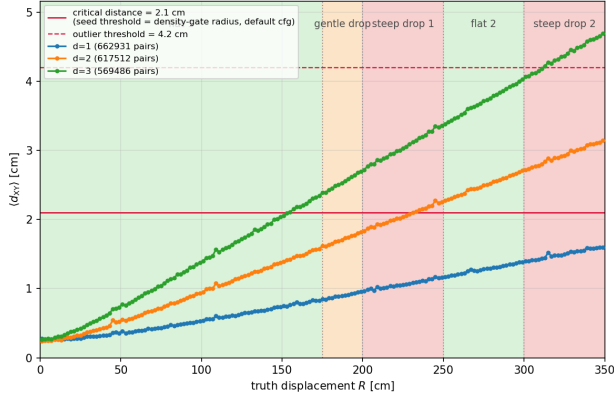


(a) Full projected distance for core-LC pairs

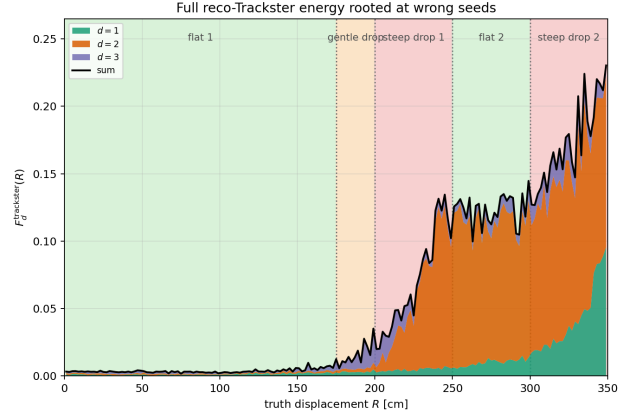


(b) Reconstructed energy in additional seeded fragments

Figure 8: The connection between the geometric threshold and fragmentation. (a) Mean full projected distance for LC pairs per d . (b) Full reconstructed energy in Tracksters whose seed has a higher-density candidate beyond 1.8 cm, separated by the layer gap d to that candidate.

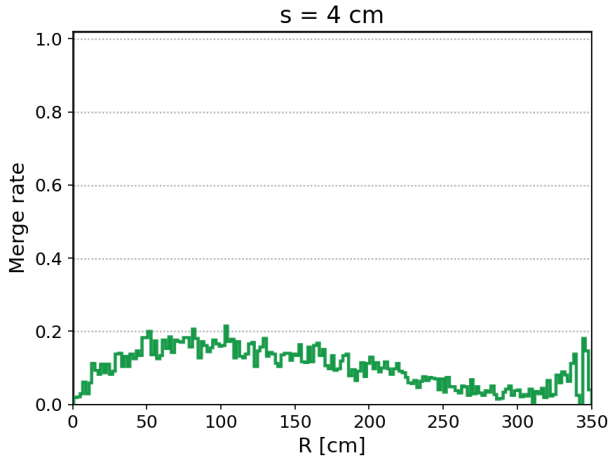


(a) Full projected distance for core-LC pairs

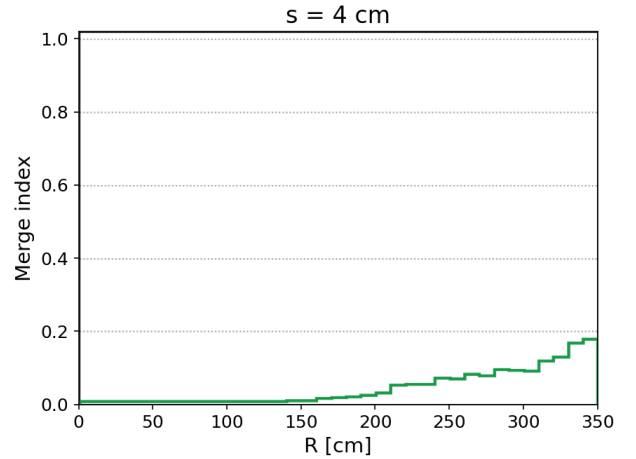


(b) Reconstructed energy in additional seeded fragments

Figure 9: The diagnostics of Fig. 8 for $d_m = 2.1$ cm.

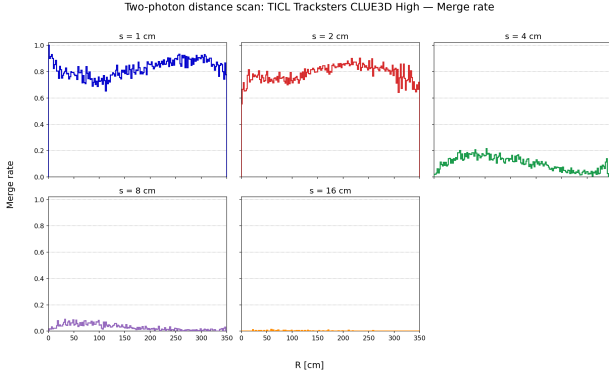


(a) Merge rate.

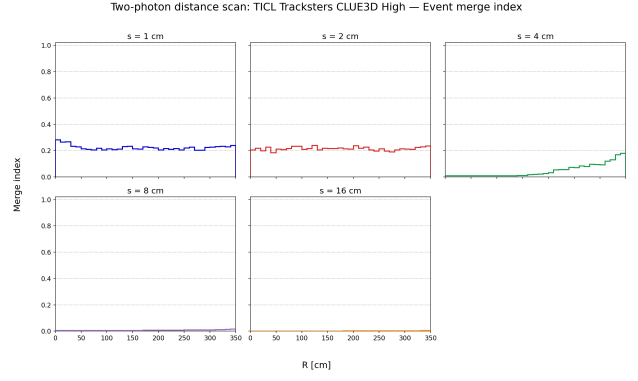


(b) Cross-shower link index.

Figure 10: Two-photon distance scan with default CLUE3D at $s = 4$ cm.



(a) Merge rate.



(b) Cross-shower link index.

Figure 11: Two-photon separation scan with default CLUE3D for $s = 1, 2, 4, 8, 16$ cm.

break an object that matched two showers into smaller pieces that no longer pass both association cuts for merge, and also increases total number of tracksters. A decreasing merge rate therefore does not establish that the algorithm has become better at separating the photons.

7.2 A diagnostic based on the clustering links

To better measure merging, I introduced a *merge index* that measures cross-shower mixing directly from the LayerCluster links, without depending on how many Tracksters are produced. Every follower LC i has a nearest-higher neighbour $h(i)$. If $h(i)$ is owned by a different truth shower than i , the algorithm has linked across two showers, which ideally should not happen. The merge index is the energy-weighted fraction of follower energy in such links,

$$I_{\text{merge}}(B) = \frac{\sum_{i \in \mathcal{F}_B} E_i [o(h(i)) \neq o(i)]}{\sum_{i \in \mathcal{F}_B} E_i}, \quad (21)$$

where $o(\cdot)$ is the LC's truth shower, E_i is LC energy, and $\mathcal{F}_B$ are follower LCs whose truth shower is in displacement bin B .

The index measures energy-weighted LayerClusters linked to wrong nearestHigher, instead of the total energy eventually assigned to the wrong shower. One crossing link can redirect a whole chain of followers, and seed and outlier decisions change which LCs enter the denominator. These differences make it complementary to the object-based merge rate. Unlike merge rate, this diagnostic is not generic: it requires a nearest-higher link structure and therefore applies only to CLUE-like algorithms.

Figure 10b denies the apparent improvement. For $s = 4$ cm, the index rises from close to zero near the origin to about 0.18 at large displacement. Cross-shower linking becomes more frequent even while the reconstructed-object merge rate decreases.

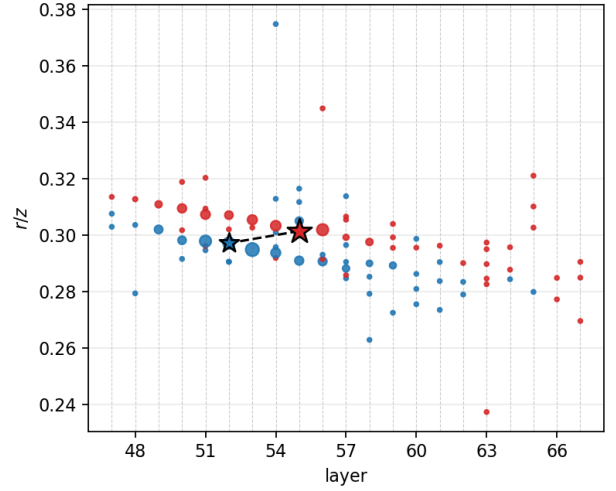


Figure 12: An actual cross-shower nearest-higher link for photons separated by $s = 4$ cm at $R \simeq 131$ cm, shown as r/z versus layer. Blue marks the simulated truth shower A and red the simulated truth shower B , rather than reconstructed Trackster membership; the dashed line is the recorded nearest-higher hop from B 's highlighted LC (layer 55) to A 's (layer 52). The two lie at nearly the same r/z despite belonging to different showers.

7.3 Connection to the projective geometry

Figure 12 shows an individual cross-shower link, in an event with $R \simeq 131$ cm. The highlighted red LC belongs to shower B , but its recorded nearest-higher neighbour is the blue LC from shower A , three layers away. The link is geometrically possible because the two LCs sit at nearly the same $r/|z|$ and ϕ despite belonging to different showers, so once projected onto the same layer the neighbour falls within the critical distance.

The origin projection discussed in Section 6 can also explain how links cross between the showers. Consider ideal parallel axes $\mathbf{r}_A(z) = \mathbf{b}_A + z\mathbf{a}$ and $\mathbf{r}_B(z) = \mathbf{b}_A + \mathbf{s} + z\mathbf{a}$, where $\mathbf{s}$ is their transverse separation. Projecting a point on shower B at z_j onto the layer of a point on shower A at z_i gives distance:

$$\mathbf{r}_A(z_i) - \frac{z_i}{z_j} \mathbf{r}_B(z_j) = \mathbf{b}_A \frac{z_j - z_i}{z_j} - \frac{z_i}{z_j} \mathbf{s}. \quad (22)$$

The first term is the previously explained projective distance error; the second represents the true separation of the pair. For suitable shower positions, these terms partly cancel. Points from different showers can then appear close in the projected plane even while points along one shower are pushed apart.

7.4 Which layer gaps produce the merging?

Because the projection offset grows with Δz , higher-order links bridge s at smaller R and cross first. Shorter-gap links $d = 1, 2$ happen as R increases.

The two-photon study therefore shows why tuning must consider both shower completeness and crossing shower links. Recovering energy in larger Tracksters alone is not sufficient.

8 CLUE3D performance with pileup

Pileup introduces many nearby showers with a broad range of energies and particle types. I studied samples with an average of 200 additional interactions per bunch crossing (PU200), keeping two efficiencies apart: the efficiency of reconstructing the injected signal photon, and the inclusive efficiency measured over all selected in-time² SimTracksters (signal plus pileup). Signal-only curve tests the same displaced photon topology in a busier environment. Inclusive curve also changes the population being measured.

8.1 Signal efficiency and the meaning of displacement

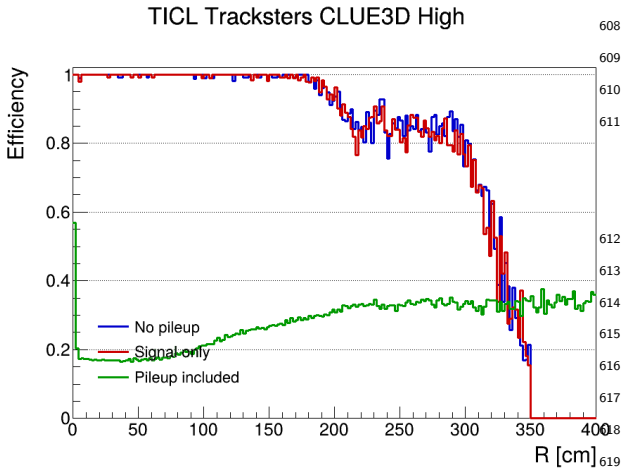


Figure 13: Default CLUE3D efficiency without pileup, for the generated signal photon in PU200, and for the inclusive in-time truth selection in PU200.

Figure 13 compares the no-pileup photon sample, the signal photon in PU200, and the inclusive PU200 selection. The first two curves are similar: both have high

²Produced in the current bunch crossing, rather than in the previous or following bunch crossings.

efficiency at small displacement and show the large- R degradation identified in Section 6. The inclusive curve is quite different, and it is dominated by pileup: the signal photon accounts for fewer than 1 in 1000 of SimTracksters, so the inclusive denominator is essentially the pileup population. It falls sharply just outside the first displacement bin, reaches a minimum, and then increases over a broad range of R .

For pileup particles, resolved R is particularly easy to misinterpret. The HGCal boundary position and direction define a line that can be extrapolated to $z = 0$. Charged particles bend in the magnetic field, so this line is a tangent to a curved trajectory. Its R can be far from 0 even when the particle was produced close to the IP. The reconstruction consequences, however, are the same either way: a trajectory with a large extrapolated R still projects poorly onto the interaction point, so by Eq. (19) its LayerClusters are pushed apart as $d_{XY} \simeq R|\Delta z|/z$ and suffer the same fragmentation as a genuinely displaced shower.

Overall, adding PU200 leaves the signal-photon efficiency largely unchanged: the loss at large R persists, while the inclusive curve mainly describes the reconstruction of pileup particles.

8.2 Separating particle composition from reconstruction effects

To investigate the inclusive shape, I broke the efficiency down per particle in a 20 000-event study.

The truth composition changes rapidly near $R = 0$. Photons contribute strongly in the first bins since they are not curved by magnetic field, while charged pions dominate most of the larger resolved-displacement range (Fig. 14). This accounts for part of the change, but it is not a complete explanation: efficiencies within individual species still depend on R .

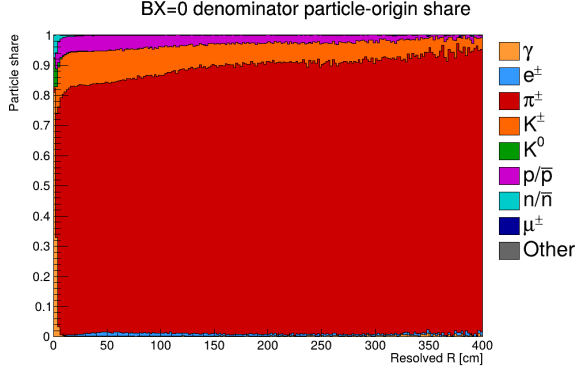
Even for a straight photon trajectory, the longitudinal spread of collision vertices matters. For a PU photon produced on the beamline at $z = z_v$, the transverse intercept of its line at $z = 0$ has magnitude

$$R = |z_v| \tan \theta, \quad (23)$$

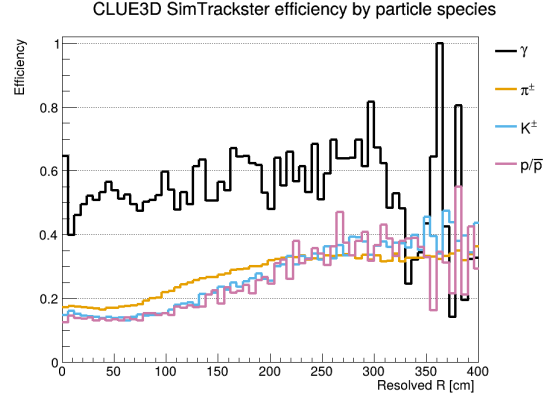
where θ is the momentum polar angle relative to the beam axis.

For photons, dependence at small R is angular rather than on displacement. Pileup photons come from vertices close to the beamline, so z_v is small and Eq. (23) makes resolved $R = |z_v| \tan \theta$ essentially a proxy for polar angle. A two-dimensional efficiency map in $(R, |\eta|)$ confirms this, with the structure changing in $|\eta|$ rather than R . Loss is present in the lower- $|\eta|$ edge of the HGCal acceptance; a one-dimensional photon-versus- R curve inherits that acceptance edge and only looks like a displacement effect.

For charged pions, kaons and protons the efficiency rises with resolved R , and the explaining variable is the LC multiplicity of the truth shower: showers with many LCs rarely have more than half of their energy collected in a single Trackster. As R grows the



(a) Composition of the in-time truth denominator.



(b) Efficiency split by particle species.

Figure 14: Particle composition and reconstruction efficiency versus resolved R in pileup.

whole pion LC-multiplicity distribution shifts downward, so the population moves towards the smaller, few-LC showers that CLUE3D reconstructs more easily, which was not fully explained why. Higher-momentum hadrons bend less, keeping a smaller resolved R while developing larger showers, so larger R preferentially selects the lower-multiplicity ones.

Inclusive pileup efficiency must therefore be read together with particle identity, energy, angle and shower size, and the signal-only selection remains the test of displaced-photon reconstruction in pileup environment.

the reach per axis,

$$D_{2,w} = \sqrt{\sum_k w_k (\Delta x_k)^2}, \quad (27)$$

$$D_{1,w} = \sum_k w_k |\Delta x_k|, \quad (28)$$

$$D_{\infty,w} = \max_k (w_k |\Delta x_k|), \quad (29)$$

so a larger w_z shortens the longitudinal reach. All these metrics depend only on coordinate differences and make no projection towards the interaction point.

9 Comparison with CLUEstering

CLUEstering is a generalized, heterogeneous implementation of the CLUE approach, with configurable coordinates and distance metrics, meaning projective coordinates are only one of the options. It retains the density, nearest-higher and seed-follower structure introduced in Section 5. My work was to run and diagnose its application to displaced showers in CMSSW, comparing several configurations with default CLUE3D.

9.1 Cartesian distance metrics

For Cartesian coordinates (x, y, z) I tested three distance families. Writing $\Delta x = x_i - x_j$, and similarly for y and z , their unweighted forms are

$$D_2 = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}, \quad (24)$$

$$D_1 = |\Delta x| + |\Delta y| + |\Delta z|, \quad (25)$$

$$D_\infty = \max(|\Delta x|, |\Delta y|, |\Delta z|). \quad (26)$$

A fixed cut selects a sphere for the Euclidean distance D_2 , an octahedron for the Manhattan distance D_1 and a cube for the Chebyshev distance D_∞ . Positive weights w_k on the coordinate differences Δx_k adjust

9.2 A projective metric under development

I also tested a work-in-progress projective configuration, labelled *Gnomonic* in the plots. It maps LC positions to the direction coordinates

$$\mathbf{g} = \left(\frac{x}{|z|}, \frac{y}{|z|} \right). \quad (30)$$

It uses the layer index ℓ as the longitudinal coordinate, and applies the cylinder distance

$$D_{\text{cyl}}(i, j) = \max \left(\frac{|\mathbf{g}_i - \mathbf{g}_j|}{\sigma_T}, \frac{|\ell_i - \ell_j|}{L} \right). \quad (31)$$

Pointing showers have constant $\mathbf{g}$; non-pointing ones drift across this plane, so some degradation with displacement is still expected. The tested implementation clusters only the electromagnetic section (EE), with $\sigma_T = 0.003$, $L = 4$ layers, density radius $d_c = 1$, seed distance 2.8 and outlier distance 3.6 in the normalized units of Eq. (31). Its density threshold varies with direction to match the η -uniform pileup density,

$$\rho_c(\mathbf{g}) = 0.8 \text{ GeV} \left(\frac{0.42}{|\mathbf{g}|} \right)^2. \quad (32)$$

9.3 First comparisons without pileup

These are exploratory tests, not the result of tuning. For the Cartesian configurations I held the density radius at $d_c = 5 \text{ cm}$ and the density threshold

at $\rho_c = 6$ GeV and varied the nearest-higher distance d_m , identified in the previous studies as the most direct control of fragmentation. The sample is 20 000 single-photon events with energies of 10–250 GeV and $R = 0$ –350 cm in Run4D121 geometry, subject to the gun’s surface-acceptance requirements.

The 6 cm Cartesian configurations largely avoid the severe large-displacement loss of CLUE3D (Fig. 15). The gnomonic configuration degrades at large displacement, as expected for projective coordinates, but fails less than CLUE3D, so it too looks promising—origin projection does not imply identical performance since the cylinder neighbourhood, distance scales and variable density all differ. The surprise is the smaller Euclidean setting: weighted 3 cm loses almost all efficiency at approximately 300 cm. Since it uses Cartesian coordinates, this failure cannot be attributed to an origin projection.

9.4 Understanding the weighted Euclidean 3 cm failure

To understand why, I plotted coordinate components of the nearest-higher links and the photon angle (Fig. 16). For a layer gap Δz , transverse step is

$$\Delta r_{xy} \simeq |\Delta z| \tan \theta, \quad (33)$$

where θ is the photon polar angle. The layer spacing makes almost constant the longitudinal part of each link, while a larger θ adds transverse distance; panel (a) shows exactly this, the transverse components growing with R while the longitudinal one barely changes. Once the full distance outgrows the 3 cm reach the link is cut and shower is split. The failure is thus not inherent to displaced particles but to large- θ ones, which displaced particles tend to be.

Clustering acts on individual links, so an average reach under the 3 cm threshold can still break many links above it. Below about 100 cm the mean photon angle falls, yet the most inclined fifth of the photons moves to larger θ (panel b).

The lesson is that Cartesian coordinates remove the artificial projective separation, but the distance reach must still be large enough to follow an inclined shower from one layer to the next.

9.5 Performance in pileup

The PU200 comparison uses a 10 000-event sample for each tested configuration. As in Section 8, I measured the generated signal photon separately from the inclusive in-time truth population.

For the signal photon, the no-pileup ordering survives (Fig. 17a): the 6 cm Cartesian configurations stay efficient across the scan, the gnomonic configuration again beats CLUE3D at large displacement, and the weighted Euclidean 3 cm failure is still present.

The inclusive picture is different (Fig. 17b). The 6 cm Cartesian metrics remain competitive—unweighted Euclidean improves on CLUE3D over most

of the range and weighted Chebyshev is broadly comparable—while Manhattan and the 3 cm choices recover much less. The gnomonic configuration is the outlier: even though its signal efficiency is the higher of the two, it collects only a small fraction of the inclusive population, several times less than CLUE3D does.

This is unlikely to be the projective metric itself: its direction-dependent density threshold probably suppresses much of the pileup, and it does not yet cluster the hadronic section, which matters for PU. Both remain to be confirmed.

The duplicate rates in Fig. 18(a) show how completely the signal is collected into one object, although one shower fragmented into many small tracksters is not considered duplicate due to threshold.

The reconstructed-object merge curves in Fig. 18 are highest for CLUE3D.

9.6 Two nearby photons and outlook

A dedicated two-photon comparison at 4 cm separation tests nearby showers directly (Fig. 19). Here efficiency cannot be read on its own: two overlapping photons collected into a single object still count as reconstructed, so a high efficiency can just as easily mean good recovery or outright merging. On that basis the gnomonic configuration holds up well: at large displacement it keeps higher efficiency than CLUE3D while its merge rate stays low and close to CLUE3D’s. It recovers more of each photon without merging the two together.

Unweighted Euclidean at $d_m = 6$ cm reaches near-perfect efficiency, but its merge rate is very high: the wide reach simply clusters both showers into one object, which is what made it look so strong on single photons. More surprising is $d_m = 3$ cm, whose short reach fragmented a single shower earlier around 0.8 efficiency: with two photons its efficiency climbs back to near 1 while it merges almost as heavily as the 6 cm case. The likely explanation is that activity from the neighbouring shower bridges the spacing a 3 cm reach cannot span alone and increases density, reconnecting the fragments into one merged object, though this would need a dedicated study to confirm.

As shown in Section 7, fragmentation can also hide mixing inside an object-based merge rate, so extending the cross-shower LC link diagnostic to this experiment would test merging more directly. Taken together, the results show that CLUEstering offers several useful choices: the gnomonic configuration improves on CLUE3D in both single- and two-photon tests, while the Cartesian metrics buy their high efficiency partly by merging and must be tuned against that. The next step is to tune coordinates, metric weights, distance reaches and density settings together, using signal and PU efficiency, duplicate, merge and fake rate to guide the choice.

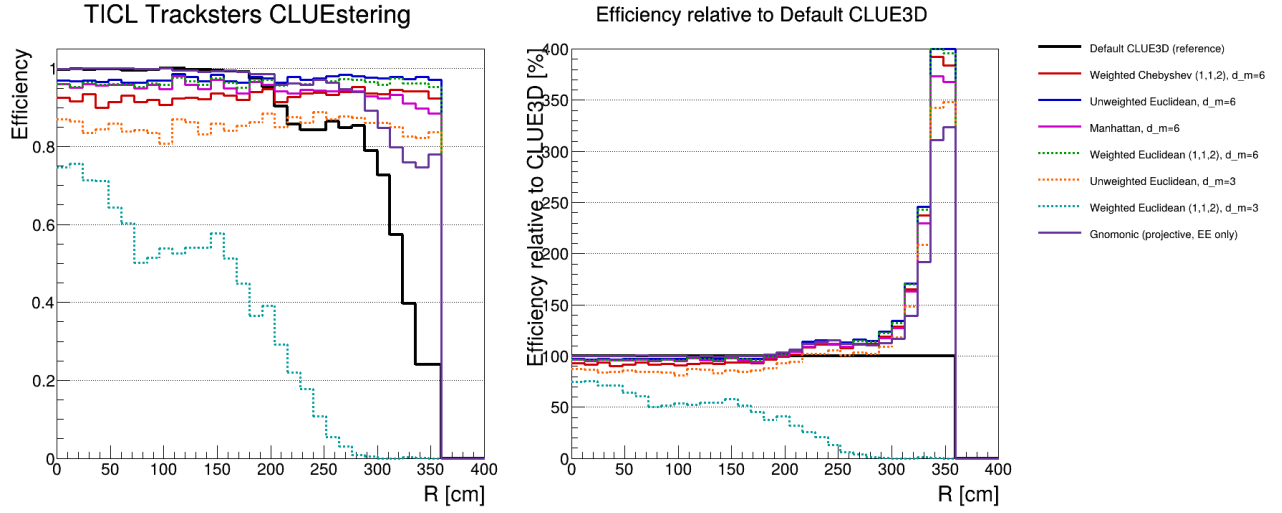
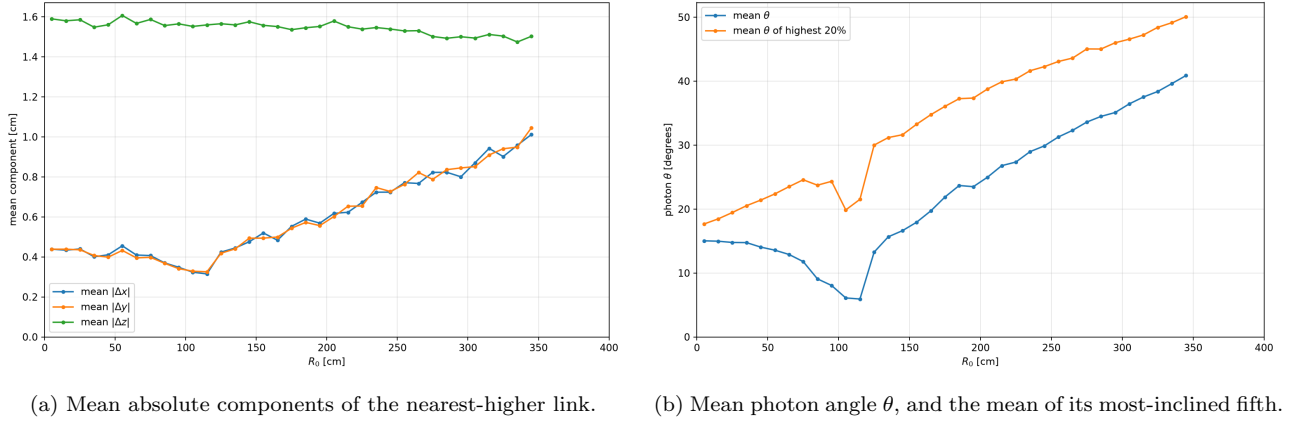


Figure 15: Single-photon efficiency without pileup for the exploratory CLUEsterng configurations. The right panel shows $100\epsilon/\epsilon_{\text{CLUE3D}}$; 100% denotes equal efficiency.



(a) Mean absolute components of the nearest-higher link.

(b) Mean photon angle θ , and the mean of its most-inclined fifth.

Figure 16: Diagnostics for the weighted Euclidean metric at $d_m = 3$ cm, versus truth displacement R_0 .

10 Conclusions and next steps

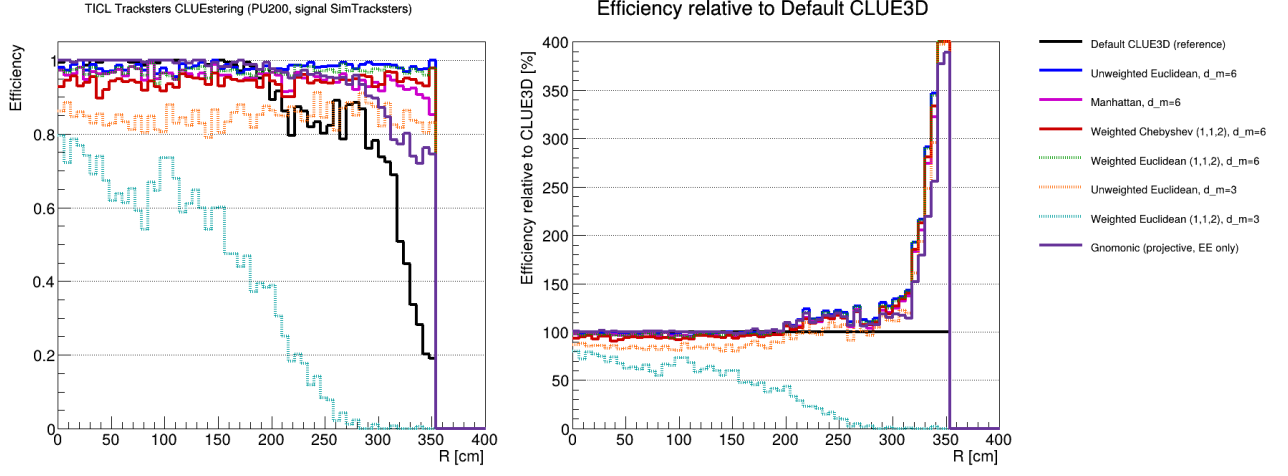
This project studied how well CLUE3D reconstructs displaced photons and why its performance changes when their showers do not point back to the interaction point. Along the way I improved the displaced-particle gun, added displacement observables to the CMSSW validation, and produced the two-dimensional performance maps used throughout. Comparing shower direction estimators gave an approximately 10 cm resolution on the back-extrapolated displacement of the most energetic Trackster. The default PCA estimator already available in CMSSW achieved the best resolution; several alternative methods reached comparable performance, but none improved upon it.

A large part of the work went into explaining why default CLUE3D loses efficiency on single displaced photons. Because it projects everything from the origin, two points on the same shower axis look transversely separated by roughly $R|\Delta z|/z$, and this separation grows with displacement. The loss comes in two stages: in the first, nearest-higher link 2 layer away breaks, so efficiency decreases and then plateaus, and at larger displacement one-layer breaks take over and drive the final decline. Widening the seed distance from 1.8 cm to 2.1 cm, pushes the fragmentation out

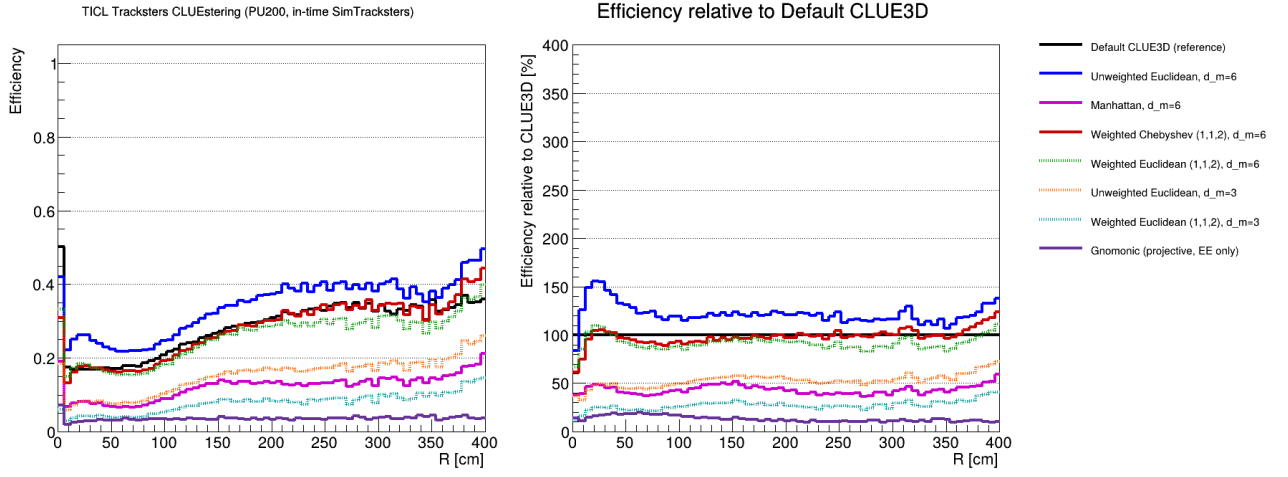
to larger displacement confirming the theory.

The two-photon studies make the case that energy recovery cannot be judged on its own. For photons 4 cm apart, the energy linked across the two showers keeps rising with displacement even where the object-based merge rate is falling, so fragmentation can make the merge rate look better while the mixing underneath is getting worse. In PU200 the injected signal photon behaves much as it does without pileup. The inclusive truth population is a different story: its particle mix, kinematics and shower sizes all shift with resolved displacement, and charged particles bend in the magnetic field, so its displacement R cannot be read as a production radius.

Comparing metrics through CLUEsterng showed both promise and limits. Several Cartesian configurations stay efficient on single displaced photons across the whole scan, including in pileup, but the Euclidean ones merge the 4 cm photon pair heavily. The tested gnomonic configuration improved efficiency over default CLUE3D at large displacement while keeping the two-photon merge rate low. Further tuning is needed before deciding which approach performs best overall. Its EE-only coverage may also contribute to its low inclusive pileup efficiency.

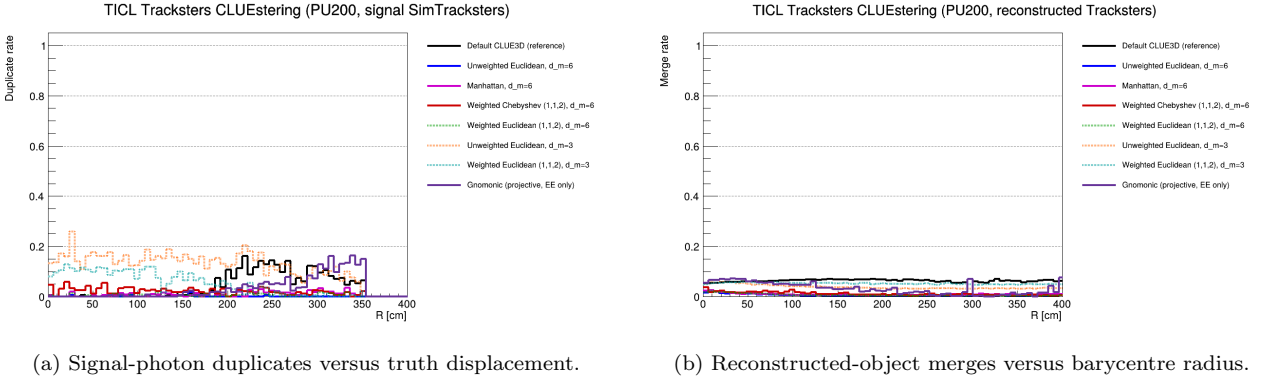


(a) Efficiency for the generated signal photon in PU200.



(b) Efficiency for the inclusive in-time truth population.

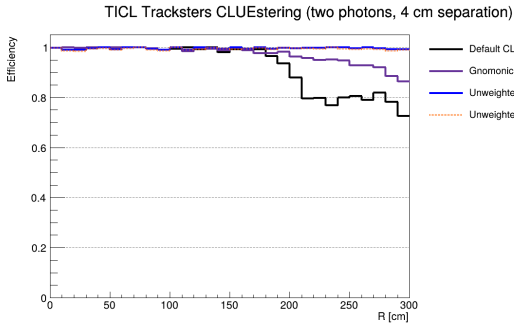
Figure 17: PU200 efficiencies, with default CLUE3D as the baseline and the same relative-efficiency convention as Fig. 15.



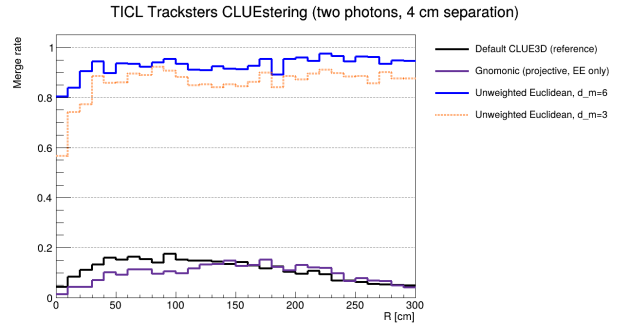
(a) Signal-photon duplicates versus truth displacement.

(b) Reconstructed-object merges versus barycentre radius.

Figure 18: Complementary PU200 rates. The duplicate denominator contains signal SimTracksters, while the merge denominator contains reconstructed Tracksters.



(a) Photon reconstruction efficiency.



(b) Reconstructed-object merge rate.

Figure 19: Two photons separated by $s = 4$ cm.

The natural next step is to tune coordinates, metric weights, distance scales and density thresholds together, judging them against energy recovery, fragmentation and cross-shower mixing at once. CLUEstoring also needs to be tested on more realistic inputs: the controlled photons here are injected just before HGCAL, so they say nothing about real LLP decays. The generation and validation tools built for this study give a starting point for that work. We would also need to establish with theory colleagues the maximum extrapolated displacement expected for relevant LLP scenarios. Since default CLUE3D performs well up to approximately 100 cm for current params in this study, it may already be sufficient.

Acknowledgements

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